FIKR Model — Lesson Plan

Learning Leading to Production

Title	Circles	**Subject / Unit**	Topic 9 — Conic Sections · Lesson 9-2
Grade	Grade 11 — Algebra II	**Date**	Week 10 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

Not shown — the Scope and Sequence document has no Essential Question column, so it is left open rather than invented. Add it when confirmed.

Lesson Objectives (curriculum map — verbatim)

1. Derive the equation of a circle given its center and radius.
2. Complete the square to find the center and radius of a circle.
3. Determine the relative positions of a circle and a line in the plane.
4. Solve a system of equations with one linear equation and one equation of a circle algebraically and verify the solution by graphing.

Standards (curriculum map — verbatim)

HSA.SSE.B.3 Choose and produce an equivalent form of an expression to reveal and explain properties of the quantity represented by the expression.

HSA.REI.C.7 Solve a simple system consisting of a linear equation and a quadratic equation in two variables algebraically and graphically.

HSG.GPE.A.1 Derive the equation of a circle of given center and radius using the Pythagorean Theorem; complete the square to find the center and radius of a circle given by an equation.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: Center ; Circle ; Diameter ; radius.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation Before Class

Learning Outcome: What will the student be able to do?

Students will derive the equation of a circle from its center and radius using the distance formula (objective 1; HSG.GPE.A.1).

Students will complete the square to find the center and the radius of a circle given by an equation (objective 2; HSA.SSE.B.3).

Students will determine whether a line misses, touches or cuts a circle (objective 3).

Students will solve a line-circle system algebraically and verify the solution by graphing (objective 4; HSA.REI.C.7).

Three teaching days this week. If time runs short, cut Smart Production — never the Mastery Gate.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Pre-session home journey (already issued, four slides): watch "How To Find The Center and Radius of a Circle", study one worked example for $x^2 + y^2 - 6x + 4y - 12 = 0$, answer four ungraded readiness questions. Submitted on the LMS before class.

Readiness questions: the center and radius of $(x + 1)^2 + (y - 4)^2 = 49$; the center and radius of $x^2 + y^2 + 8x - 2y + 8 = 0$; the equation with center $(2, -3)$ and radius 4; the equation with center the origin through $(3, 4)$.

Evidence: the LMS answers feed the Diagnose.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — the slide-4 checks: complete the square for $x^2 + 6x$; find the distance from $(0, 0)$ to $(3, 4)$; solve $x^2 - 2x - 8 = 0$.

Answers: $(x + 3)^2 - 9$; 5 ; $x = 4$ or $x = -2$.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gaps — Q1: the -9 is left out. Q3: the roots are given with the wrong signs.

Routing: 0–1 correct → re-teach completing the square and factoring. 2 → straight to the circle equation. 3 → begin Investigate early.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 5 — The equation of a circle (objective 1; HSG.GPE.A.1). $(x - h)^2 + (y - k)^2 = r^2$ comes from the distance formula; center $(3, -2)$, $r = 5$ gives $(x - 3)^2 + (y + 2)^2 = 25$; center $(-4, 1)$ through $(-1, 5)$ has $r^2 = 25$.

Slide 6 — Completing the square (objective 2; HSA.SSE.B.3). $x^2 + y^2 - 6x + 4y - 12 = 0$ becomes $(x - 3)^2 + (y + 2)^2 = 25$; $x^2 + y^2 + 8x - 2y + 8 = 0$ has center $(-4, 1)$, $r = 3$; $2x^2 + 2y^2 - 12x + 8y - 8 = 0$ has $r = \sqrt{17}$.

Slide 7 — A line and a circle (objective 3). With $x^2 + y^2 = 25$: $y = 3$ cuts at $(\pm 4, 3)$; $y = 5$ touches at $(0, 5)$; $y = -7$ misses. Compare the distance d from the center with r.

Slide 8 — Solving the system (objective 4; HSA.REI.C.7). $x^2 + y^2 = 25$ with $y = x + 1$ gives $(3, 4)$ and $(-4, -3)$;

$(x-2)^2+(y+1)^2=8$ with $y=x-3$ gives $(0,-3)$ and $(4,1)$; $y=3x+10$ is tangent to $x^2+y^2=10$ at $(-3,1)$.

Narration focus — the number on the right is r^2; flip the signs for the center; a repeated root means a tangent.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: the circle $(x+1)^2+(y-2)^2=36$ — write the center and the radius.

Expected: center $(-1,2)$, radius 6. Watch for $(1,-2)$ and for radius 36.

80% correct → release guided practice.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two identical "Give it a go" problems, worked with the teacher:

(1) Find the center and radius of $x^2+y^2-10x+6y+18=0$. (2) Write the equation with center $(0,3)$ and radius $\sqrt{7}$.

Answers: (1) $(x-5)^2+(y+3)^2=16$, center $(5,-3)$, $r=4$. (2) $x^2+(y-3)^2=7$.

Worked live; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

"Choose Your Route" sheet with live Pear Deck feedback. Routes name the task, not the student.

PRACTICE — Secure the method. Six cards: read the center and radius, write the equation, complete the square; then find the error and sketch one circle on the grid. Done when: the center is read with the signs flipped and the right-hand side is treated as r^2.

APPLY — Use it in context. A plaza and a path, a radio tower, a lake and a path. Done when: the center is placed first and each answer says what it means.

INVESTIGATE — Find out why. Derive the equation, show a tangent by a repeated root, and prove the tangency condition $c^2=r^2(1+m^2)$. Done when: the argument starts from the distance formula or a repeated solution.

Page 3: compare three routes, judge three claims, test a What if...? idea. Every student sits the same Mastery Gate.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 11).

A model circular reflecting pool in the Hira Cultural District has the equation $x^2+y^2-8x-6y=0$ (metres). The model values are invented for practice.

(a) Find the center and radius and show the pool passes through the origin. (b) A walkway along $y = x$ crosses the pool: find where it enters and leaves, and the length over water. (c) A walkway along $y = 8$: does it miss, touch or cut the pool?

Answers: (a) $(x-4)^2 + (y-3)^2 = 25$, center $(4, 3)$, $r = 5$. (b) $(0, 0)$ and $(7, 7)$, length $7\sqrt{2} \approx 9.9$ m. (c) distance from the center $= 5 = r$, so it touches at $(4, 8)$.

Students then hand their reasoning to the AI challenger.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum standard: standard form, center and radius, both intersection points, units (m).

Role of AI: challenger only — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently: the written justification for (a)–(c).

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Write the equation with center $(-3, 2)$ and radius 4. 2. Center and radius of $x^2 + y^2 + 6x - 4y - 12 = 0$. 3. Solve $x^2 + y^2 = 20$ with $y = x + 2$.

Answers: 1. $(x+3)^2 + (y-2)^2 = 16$. 2. $(x+3)^2 + (y-2)^2 = 25$: center $(-3, 2)$, $r = 5$. 3. $(2, 4)$ and $(-4, -2)$.

Done when: question 2 shows the completing-the-square line and question 3 gives both points.

Scored live in Pear Deck and logged in the Impact Dashboard.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise their pool reasoning using the AI challenger's feedback and the teacher's micro-conference note.

Final product (Canva one-pager, photographed solution, or 45-second voice note): the equation in standard form, the walkway intersections, ONE non-example (radius read as r^2), and why a repeated root means a tangent.

Graded on clarity of reasoning.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall.

Total minutes across the seven stages: **60** (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Six cards, a find-the-error line and a sketch. Done when: the center is read with the signs flipped.

APPLY — Use it in context. A plaza, a tower, a lake and a path. Done when: the center is placed first.

INVESTIGATE — Find out why. A derivation, a tangent and the tangency condition. Done when: the argument starts from the distance formula.

No student is labelled; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Hira Cultural District plaza and pool, a radio tower and a lake; invented values.

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Geometry and Trigonometry: $x^2 + y^2 - 6x + 8y = c$ is a circle of radius 7, so $c = 24$. Trap: using $c = 49$.

GAT (Qudurat) — the radius of $x^2 + y^2 - 4x + 6y - 3 = 0$ versus 3: $r = 4$, so the radius is greater. Trap: taking $r^2 = 16$ as the radius.

SAAT (Tahsili) — the line $y = x + k$ with $k > 0$ is tangent to $x^2 + y^2 = 8$ when $k = 4$. Trap: $k = \sqrt{8}$.

Suggestions for Improvement: